

MGT 816 · The Private Firm CFO · Arc 3 · Plan & Measure · Session 6 of 12

KPIs, Budgeting & Forecasting

Driver-Based Planning and Unit Economics

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Wednesday, September 23, 2026

**PS1 due today 1:00 pm · Case 2 memo (Solstice Robotics) due Wed 9/30 ·
Laptops required Monday.**

Did You Know? Euro Disney ★

DID YOU KNOW?

“US Disney parks were instant hits. Euro Disney nearly collapsed. Why?”

1986: Disney built Euro Disney projections from its other parks' data.

US playbook: start prices low, raise slowly, let early adopters spread the word.

Euro Disney assumed aggressive attendance at a premium price - both at once.

By 1993 it slashed ticket prices to fill the park; no word-of-mouth buzz.

**The arithmetic was fine.
The assumptions were
wrong.**

**AI makes the arithmetic
free. Assumptions are still
your job.**

Roadmap: today's six topics

1. The plan (the AOP)

2. Drivers, not line items

3. The feedback loop

4. KPIs that matter

5. Unit economics in 60 seconds

6. Prune an AI's KPI tree

Why plan? \$25,000 and four ways to spend it

New startup, \$25,000 in the bank. Where does it go?

product development

marketing

a sales force

More marketing → will the extra sales offset the cost?

+25% engineering (~10 people) → does that trigger another financing round?

Budgeting = evaluating trade-offs + building the machinery to respond when reality differs.

The AOP: the plan you don't change ★

AOP = a 12-month financial plan; approved at the Nov/Dec board meeting.

Often the bonus contract - management comp partly tied to the plan.

Cost containment: the discipline for cash-constrained companies.

The base plan never changes mid-year - it is the benchmark.

The monthly forecast updates: closed-month actuals + projections for the rest.



Rolling forecasts + the 80-day AOP calendar: see pre-read (S9, S11).

Top-down meets bottom-up

Top-down

(from board + management)

revenue targets → profitability targets → cost constraints.

Bottom-up

(finance + function leads)

activities to hit revenue → cost of activities + headcount → profitability determined.



Finance leads the process and mediates the cross-functional fight over constrained resources.

The two-pass compile ★

Discrepancies between the board's number and the bottom-up build are normal.

Kupor (a16z), “budgeting is a two-pass compile”:

Pass 1: board hands down goals/constraints; management builds and “bursts into tears.”

Pass 2: negotiate to a plan both sides own (often a base case + stretch goals).

“Board says \$4M. Bottom-up says \$2.8M. It's Monday morning - what do you do?”

Driver-based planning: budget activities, not line items ★

Drivers = metrics informative of key business activities.

Emphasize what management **controls**.

Links every decision to strategy; focuses time on what moves results.

Enables fast “what-if” scenarios and more accurate projections.

The alternative - ask each department head to list every expense - produces a phone book, not a plan.

One month of e-commerce: the assumption table

Driver	Description	Assumption
# of impressions	“Eyeballs” that see the online ad	5,500,000
Click-through rate	% of impressions that click the ad	1%
Cost per 1,000 impressions (CPM)	Cost to reach 1,000 people	\$0.80
Conversion rate	% of clickers who purchase	5%
Average price of merchandise	Average price per unit sold	\$4.00
Average cost of merchandise	Average cost per unit sold	\$1.00

The driver chain: assumptions → contribution margin ★

CPM \$0.80 × impressions 5,500,000	=	Total marketing \$4,400
impressions 5,500,000 × CTR 1.00%	=	Website visits 55,000
visits 55,000 × conversion 5.00%	=	New customers 2,750
new customers 2,750 × price \$4.00	=	Total revenue \$11,000
new customers 2,750 × unit cost \$1.00	=	Total cost of revenue \$2,750
revenue \$11,000 – COGS \$2,750 – marketing \$4,400	=	Contribution margin \$3,850

 Driver  Outcome

Assumptions are the argument

Every driver value is an assumption (conversion 5%, price \$4...).

Take inventory of BOTH explicit and implicit assumptions.

The plan's value is the **discussion** it forces.

Open dialogue between managers and investors catches flawed assumptions.

Dangerous-implicit-assumptions list - McGrath & MacMillan, "Discovery-Driven Planning," 1995.

Drivers map to strategy questions

Interrogate the strategy model - “Does it work? Does the market care? Can we protect it?”

Strategy question	Driver	Data source	Budget line
Does the market care?	Conversion rate, units, adoption %	CRM, sales pipeline, usage data	Revenue (top line)

Two more mappings (does it work? / can we protect it?) and the directed-search example are in the appendix / notes.

The CEO & CFO dynamic ★

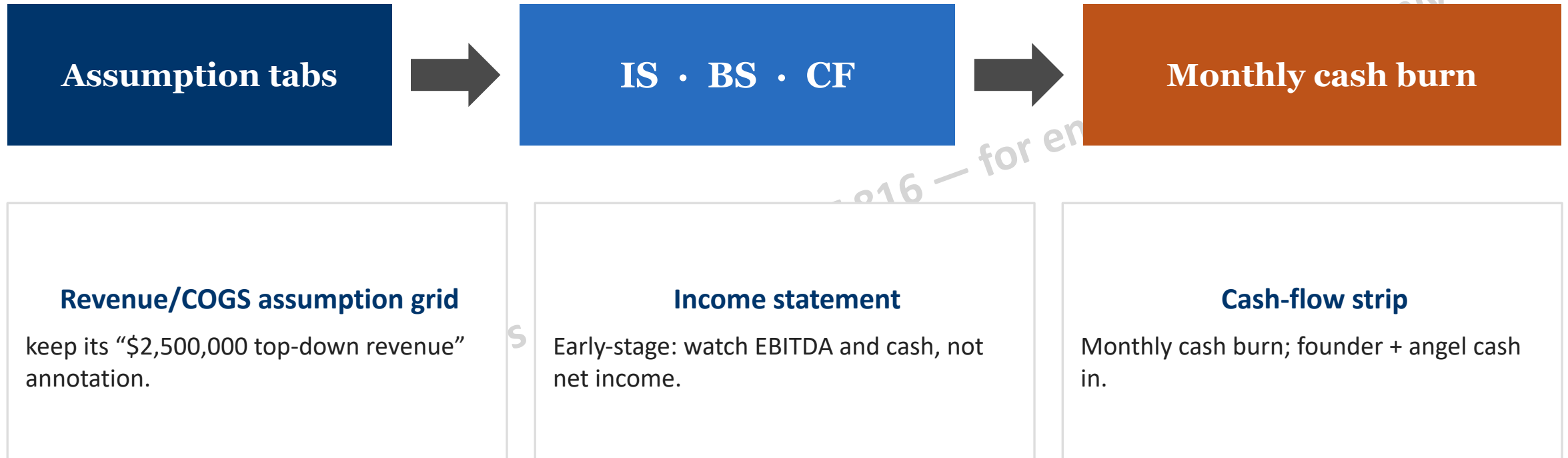
How do you tell an eternal-optimist CEO the company can't scale that fast?

Put the vision into a **driver model** - then argue about assumptions, not feelings.

A driver-based plan is what makes that conversation possible.

“Doubling sales requires an 80% higher click-through rate. Is that achievable?”

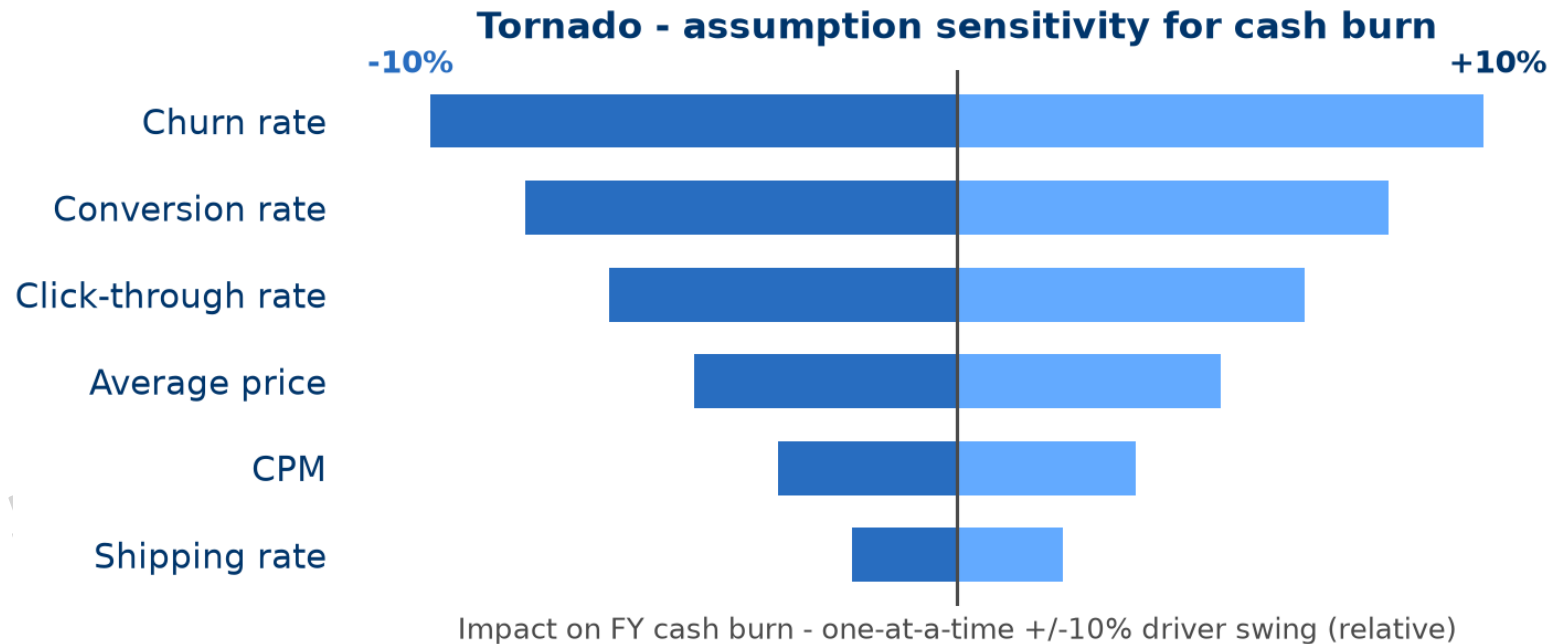
The model emits the statements - and the burn



Drivers go in; the three statements + the burn come out.

Sensitivity: which assumptions matter most ★

Small changes in some assumptions swing the whole plan. Sensitivity analysis ranks them.



Illustrative ranking; magnitudes come from the model's data table (Budget Model v1.03).

The tornado tells the CFO which drivers to monitor - it is not a forecast.

Variance: plan vs. reality ★

Variance analysis finds root cause, not blame.

It may say: revise an assumption (pricing too aggressive, CAC too conservative)...

...or fix a model gap (are ALL acquisition costs actually in CAC?).

A revenue beat can hide a margin miss.

Re-forecast: update drivers, never outputs ★

Plan and reality WILL diverge - so re-forecast.

Update the forecast's **drivers** with better-informed assumptions; let the model re-emit outputs.

Iterative: you learn which drivers predict; one driver may split into channel-level drivers.

Keep the current-year base plan static; the re-forecast feeds it forward.

**driver assumption updated
based on Q1 actuals**



[driver cell]

KPIs are cockpit instruments

KPIs give management critical information continuously - think cockpit gauges.

Critical question	KPI
How much farther can we fly?	Fuel level (→ cash runway)
How expensive is it to add a customer?	Customer acquisition cost (CAC)
Is the website selling?	Conversion rate
When do we run out of cash?	Burn rate

A good KPI set answers the questions management asks every week.

All KPIs are metrics; not all metrics are key ★

KPIs translate strategy into a coherent set of measures.

Employees focus on what's measured - so choose carefully.

KPIs change as strategy and stage change (Seed ≠ Growth); retire stale ones.

The right KPI set depends on the business model (license vs. subscription vs. usage).

All KPIs are metrics - but not all metrics are key.

Strategy → KPI cascade: a subscription example

Mission: help people live smarter by delivering high-quality personal-care products + content + technology.

Affordable, fast purchases

Tactics: fair/simple pricing, free shipping, speedy delivery, easy website

KPIs: subscription price, cost of shipping, on-time deliveries, customer-survey results

Win millennial share

Tactics: create buzz, encourage trials, build sticky relationships

KPIs: # impressions, # new customers, churn rate, CAC

Retain passionate employees

Tactics: recruit right, make work fun, reward performance

KPIs: employee turnover, employee-survey results, bonuses

A dashboard tells a story

The Daily Grind (gourmet-coffee e-commerce), March dashboard:

- ✓ Profitable revenue growth
- ✓ Website traffic beating plan
- 🚩 Conversion rate weak → investigate first-time buyers
- ✓ Retention strong - converted customers stick
- 🚩 Employee-satisfaction yellow flag → act before it worsens

Effective dashboards mix financial + non-financial and tell a story with current data.

What makes an indicator “key”

- ✓ Aligned to strategy
- ✓ Attainable (data worth its cost)
- ✓ Acute (specific, not vague)
- ✓ Accurate / predictive
- ✓ Actionable (within your control)
- ✓ Alive (evolves with the business)

Built top-down from strategy.

Few enough to avoid information overload.

A deliberate mix of internal/external and leading/lagging measures.

Mauboussin, “The True Measures of Success,” HBR, Oct 2012.

Wrong metric, wrong behavior ★

KPIs and targets are incentive contracts - think about the behavior you buy.

Silo-thinking: the metric stops at your department's wall → use cross-functional metrics.

Conflicts of interest: metrics that reward harming the customer or company.

Wells Fargo opened up to 2M fake accounts to hit unrealistic sales targets; \$110M class-action settlement (CNN, Mar 2017).

Leading vs. lagging indicators

Leading (predictive)

brand recognition · NPS · customer satisfaction ·
growth in new markets · # new patents.

Lagging (results of past actions)

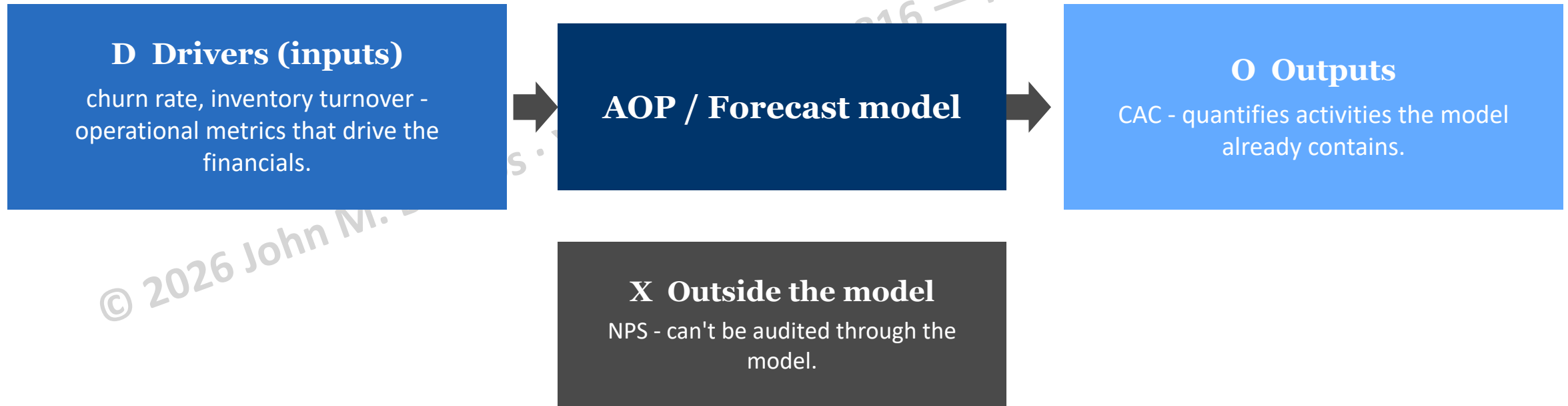
EBITDA · operating margin · cash conversion cycle.

“CEOs always act on leading indicators of good news, but only act on lagging indicators of bad news.”

- Andy Grove, former CEO, Intel.

Drivers, outputs - and everything else ★

Model emits expectation; actuals return reality; the gap is variance.



D = driver • O = output • X = outside

The unit-economics stack

CAC

Churn /
retention
(→ NRR)

LTV

NPS
(pre-read only)

How costly to get a customer · how well you keep them · what they're worth.

**You read the formulas before class. Here we work
on the judgment calls.**

CAC ★

CAC = the sales & marketing cost of convincing a customer to buy.

Varies **by channel** and across a company's life.

More spend can raise CAC; network effects / virality can lower it.

Retention is usually cheaper than acquisition (5-25x cheaper - HBR/Gallo 2014).

What vendors quote as “CAC” is often CPA - cost per acquisition step, a step short of a paying customer.

Churn: the mini-case

Lorelai & Sookie's restaurant-POS SaaS - high-touch onboarding.

Churn = % of customers (or revenue) discontinuing in a period.

Q1: 25 beginning customers, +8 new, -3 lost → 30 ending, 27.5 average.

Customer churn = $3 / 27.5 = 10.9\%$.

Poll: compute the NRR

Beginning MRR **\$100,000** (50 logos) · churned **\$6,000** · expansion **+\$8,000** · contraction **−\$2,000** · new logos **+\$9,000**.

What is net revenue retention (NRR)- and is this company healthy?

- A. 94%
- B. 100%
- C. 109%
- D. Can't tell without CAC

LTV: what a customer is worth

LTV = present value of net profit over the customer relationship.

\$200 to acquire a customer - great for an auto dealer, fatal for an e-commerce site.

customer lifetime /
retention

spend per period

gross margin

CAC

A positive LTV can still bankrupt you if CAC payback outruns cash.

LTV:CAC $\geq 3x$ - and payback pacing ★

LTV:CAC = lifetime contribution margin per acquisition dollar.

SaaS convention: $\geq 3x$ is good.

Payback period often runs **years** → pace growth to cash.

Acquiring customers faster than they pay back is how a healthy LTV:CAC still runs you out of cash.

Tunguz / NetSuite key-metric.

AARRR - and Dropbox's CAC pivot

Acquisition → site visits / CAC

Activation → account created / conversion rate

Retention → % who return / retention rate

Referral → % who refer / NPS

Revenue → spend per customer / LTV

Dropbox

Paid-search CAC > \$300 vs. an LTV that couldn't carry it.

→ killed paid search

→ freemium + referral program (April 2010: 4M users sent 2.8M referral invites)

→ **CAC solved through the funnel, not the budget.**

Exercise: Prune the AI's KPI tree

Groups of 3-4 (sit with your PS2 group). 12 minutes. Write on your printed handout - three outputs:

- (a) Cut the AI's 25 metrics to the 6 KPIs you'd put on Redline Labs' board dashboard.
- (b) Tag each of your six: D = driver of the forecast model · O = output of the model · X = outside the model.
- (c) Name one metric on the AI's list that would incentivize destructive behavior - and the behavior it buys, in one sentence.

Redline Labs - Series-A SaaS; AI contract-review copilot for mid-market law firms.

Pricing: \$2,500/mo platform fee + \$2/document above a 1,000-doc monthly allowance.

40% of firms prepay annually at a 10% discount. Support rule: 1 CSM per 15 active firms.

Biggest COGS line: AI inference cost per document (declining ~5%/quarter).

Defend any six you can trace through the model. You'll reuse your six on Monday's board one-pager - photograph your sheet.

Exhibit: the AI's 25-metric answer (unedited)

Proposed KPI framework for Redline Labs - AI-generated, reproduced verbatim.

Prompt: "You are a startup advisor. Propose the KPIs that Redline Labs - a Series-A SaaS startup selling an AI contract-review copilot to mid-market law firms at a \$2,500/month platform fee plus \$2/document usage pricing - should track. Be comprehensive."

GROWTH & REVENUE 1 Monthly recurring revenue (MRR) 2 ARR growth rate (% YoY) 3 New logos signed per month 4 Total documents processed per month 5 Average revenue per account (ARPA) 6 Net revenue retention (NRR)	SALES & MARKETING 7 Blended CAC 8 Marketing-qualified leads (MQLs) 9 Website unique visitors 10 Social-media engagement rate 11 Average sales cycle (days) 12 LTV:CAC ratio	PRODUCT & ENGAGEMENT 13 Monthly active users (MAU) 14 Net Promoter Score (NPS) 15 Average session duration 16 Model accuracy (F1) on contract clauses 17 Documents reviewed per firm per month	CUSTOMER SUCCESS 18 Active firms per CSM ("CSM efficiency - maximize") 19 Monthly logo churn rate 20 Support first-response time 21 Onboarding time-to-first-value	FINANCIAL HEALTH 22 Gross margin % (after inference costs) 23 Burn multiple 24 Months of runway 25 Revenue per employee
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Bridge: Monday you build it - with an AI analyst ★

1. You chose the dashboard; Monday you build the forecast behind it.
2. AI drafts; you verify - every number human-certified. (Euro Disney callback: arithmetic free, assumptions yours.)
3. The final exam (200 pts, closed book) tests this material without AI.

Mon 9/28, this room: laptop required (announced syllabus exception) with desktop Excel · one working LLM login per group of 3-4 - free tier explicitly sufficient · ~20-min pre-work on Canvas (Model Pack + ground rules) before class · PS2 group sign-up closes Sunday 5:00 pm · PS2 assigned Monday, due Mon 10/12 - both problem sets count (40 + 40).

Case 2 memo (Solstice Robotics) due Wed 9/30, 1:00 pm.

**Appendix - reference only, examinable at
concept level.**

Linking drivers to strategy: the directed-search example

Strategy question	Driver	Revenue model
Can we get big fast on both sides now?	Two-sided customer acquisition (buyers + sellers)	Subscription
Is peer-to-peer contact reliable?	Trust-building costs: clearinghouse, reviews, monitoring	Transaction fees
Can we monetize the connection?	Conversion of matches to paid activity	Freemium

A real portfolio-company dashboard

Hyde Park Venture Partners enterprise-software portfolio-company dashboard.

Tracks: new ARR added · contract size · rep productivity · revenue/employee · operations-staff productivity.

Actual vs. forecast by quarter, Q1 2016 - Q4 2017.

Dated 2016-17 - kept as a shape reference.

The Q1 variance + re-forecast statements

Income Statement Variance - 3 Months Ending Mar. 31, 2020 (static fallback for the live demo).

Line	Actual	AOP	Variance
Total Revenue (Q1)	\$142,313	\$124,442	Fav +\$17,871 (~+\$18K)
Gross Margin (Q1)	-43.6%	-39.5%	Unfav -4.15 pts
Full-year Forecast revenue	\$3,134,327	\$2,541,649	Fav +\$592,678

Subscriptions commentary in the sheet: "Higher than expected revenues due to lower churn rate."

Churn stabilizes with tenure: the evidence

AOL, 2014: 2.2M dial-up subscribers; the “membership” group contributed **\$143M adjusted OIBDA** vs **\$122M** for the overall business (Form 10-Q, Q1 2015).

“An increase in subscriber age tends to lead to reductions in subscriber churn” (Form 10-K, 2003).

“Our churn typically stabilizes in 8-10 quarters” (Earnings Call, Q3 2016).